Adaptive Variational Quantum Eigensolver (ADAPT-VQE)

Variational Quantum Algorithms and Many-Body Physics

Why Adaptive Ansätze?

- Standard VQE relies on fixed ansatz
- Problems:
    - Overparameterization
    - Barren plateaus
    - Poor physical interpretability
- Goal: construct ansatz *adaptively*

Physics Motivation

- Ground state problem:

$$H|\psi_0\rangle = E_0|\psi_0\rangle$$

- Variational principle:

$$E[\psi] = \frac{\langle\psi|H|\psi\rangle}{\langle\psi|\psi\rangle}$$

- Seek optimal parameterized wavefunction

Variational Quantum Eigensolver

$$|\psi(\theta)\rangle = U(\theta)|0\rangle$$

$$E(\theta) = \langle\psi(\theta)|H|\psi(\theta)\rangle$$

- Classical optimizer minimizes $E(\theta)$

Ansatz Challenge

- Hardware-efficient ansatz:
  - Shallow but unstructured
- Problem-inspired ansatz (e.g. UCC):
  - Physically motivated but deep
- Trade-off motivates adaptive methods

Coupled Cluster Ansatz

$$|\psi\rangle = e^{T}|\phi_0\rangle$$

$$T = T_1 + T_2 + \cdots$$

- T_k generates k-particle excitations

Unitary Coupled Cluster

$$|\psi\rangle = e^{T - T^\dagger}|\phi_0\rangle$$

- Hermitian generator
- Suitable for quantum circuits

Core Idea

- Build ansatz iteratively
- Select operators based on gradient
- Add most important operator at each step

Operator Pool

- Define pool:

$$\{A_k\}$$

- Example:
  - Fermionic excitations
  - Pauli strings

Gradient Criterion

$$g_k = \left|\frac{\partial E}{\partial \theta_k}\right|$$
$$= |\langle\psi|[H, A_k]|\psi\rangle|$$

Physical Interpretation

- Measures how strongly operator A_k lowers energy
- Equivalent to residual in many-body methods

Algorithm Steps

1. Initialize reference state
2. Compute gradients for all A_k
3. Select operator with largest gradient
4. Add to ansatz
5. Re-optimize parameters
6. Repeat until convergence

Ansatz Growth

$$U(\theta) = e^{\theta_n A_n} \cdots e^{\theta_1 A_1}$$